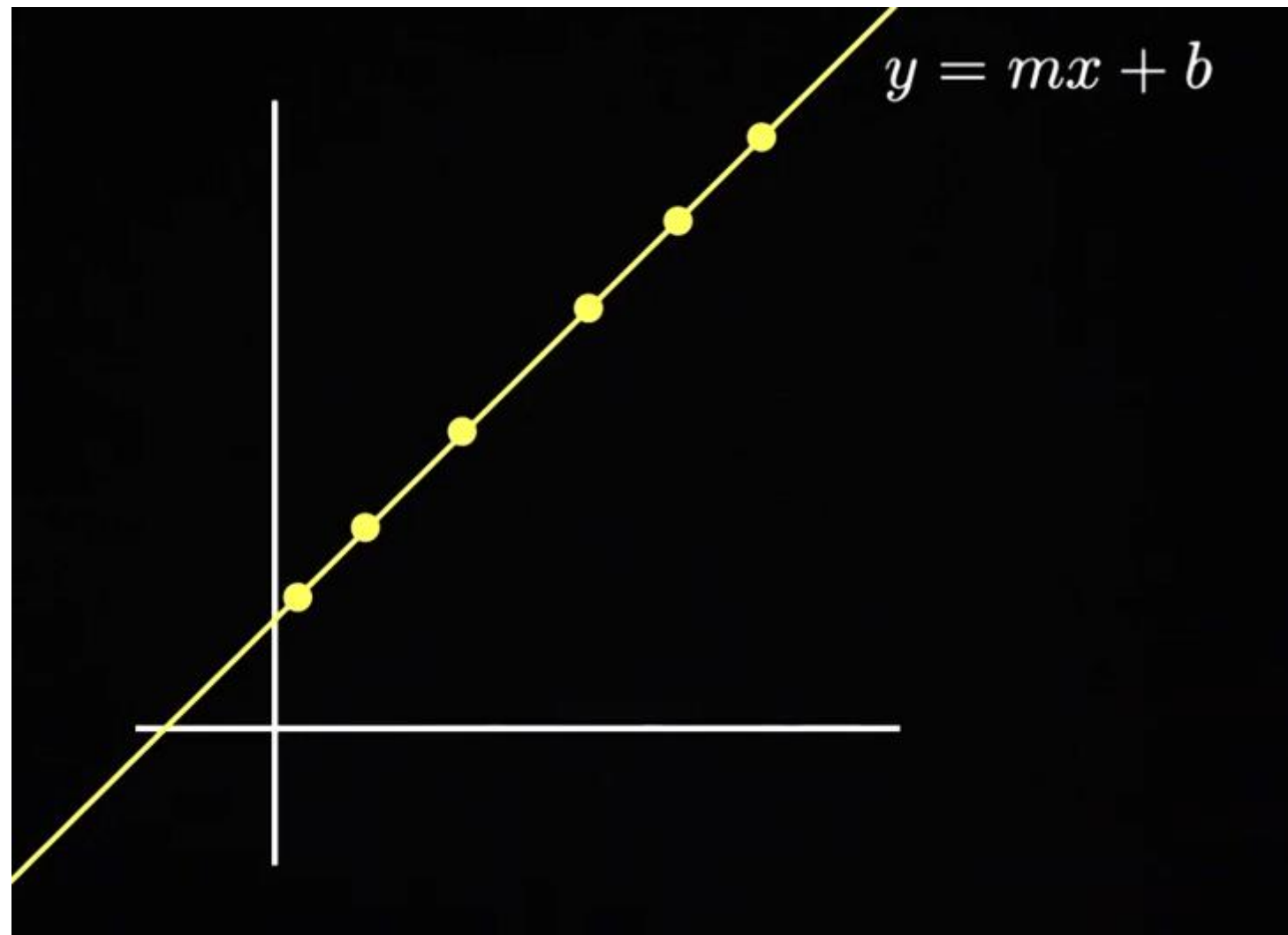
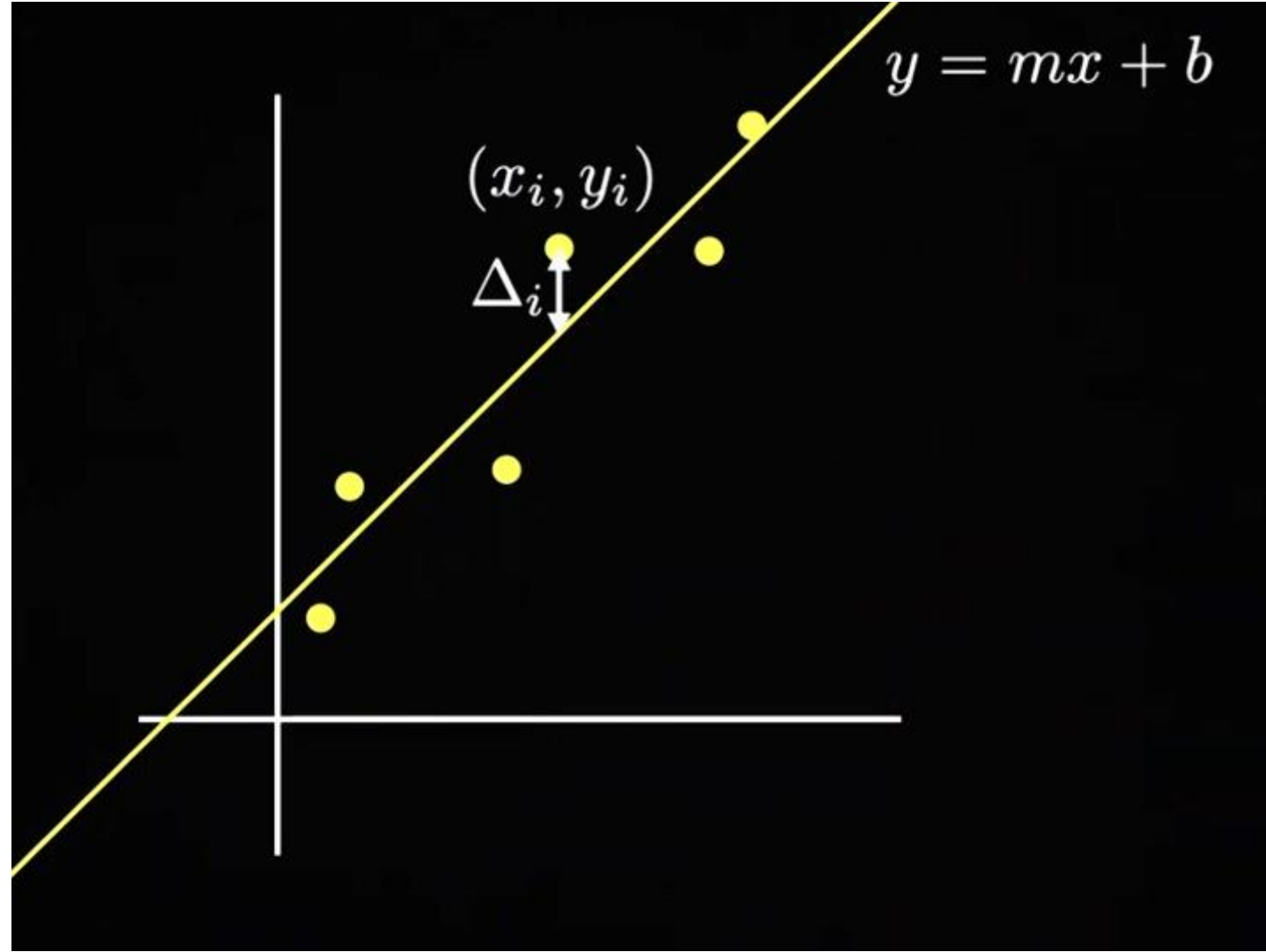
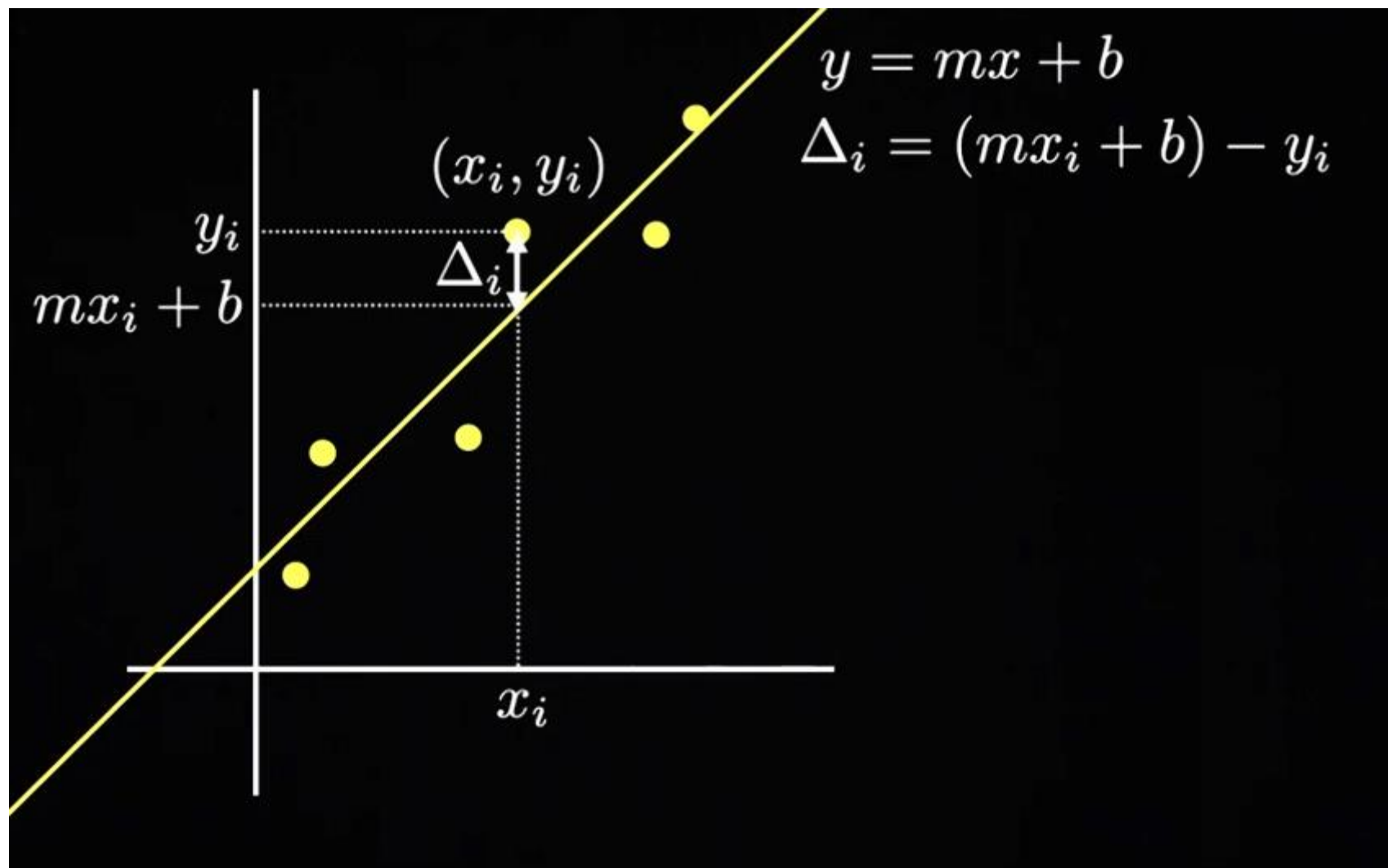
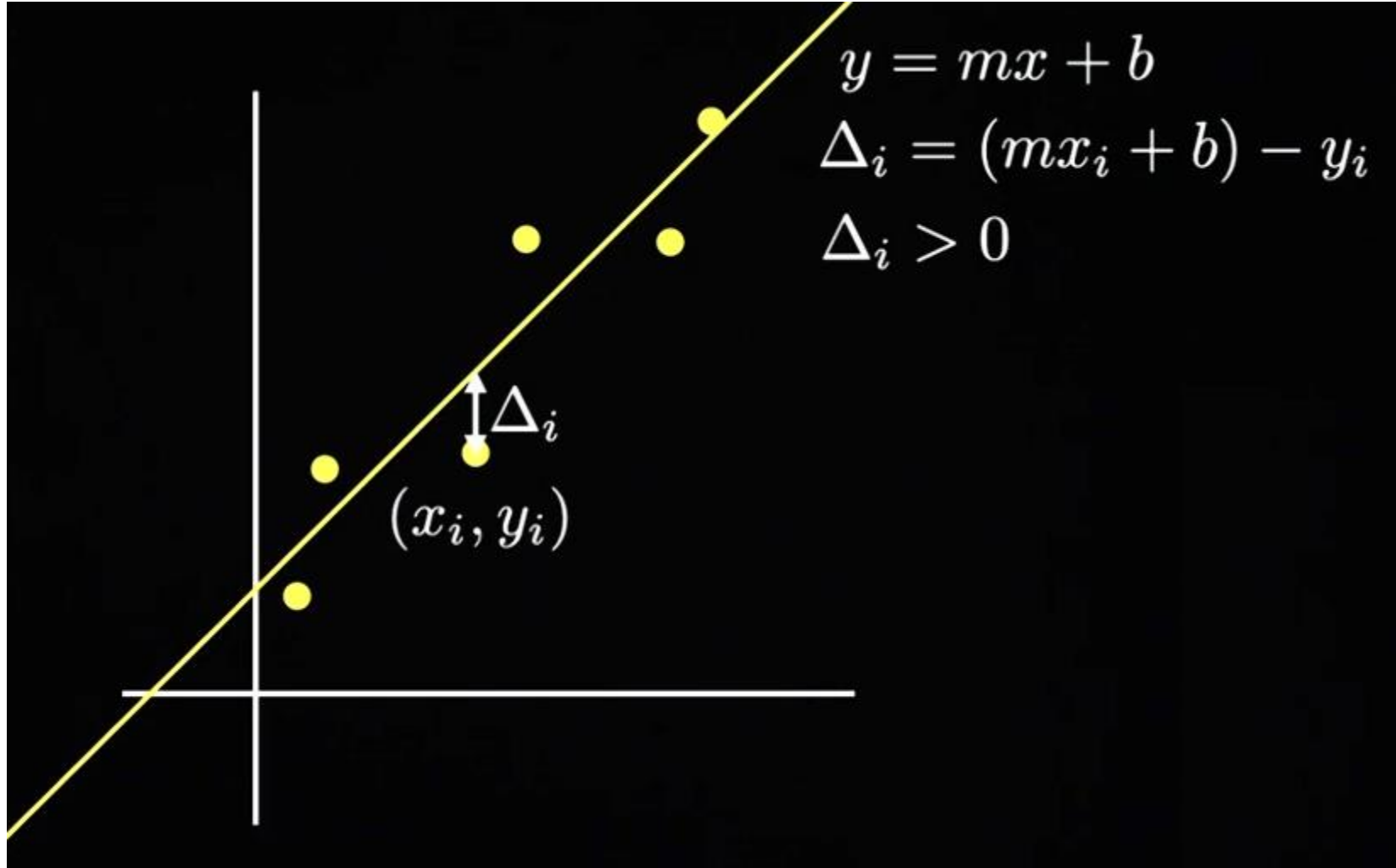


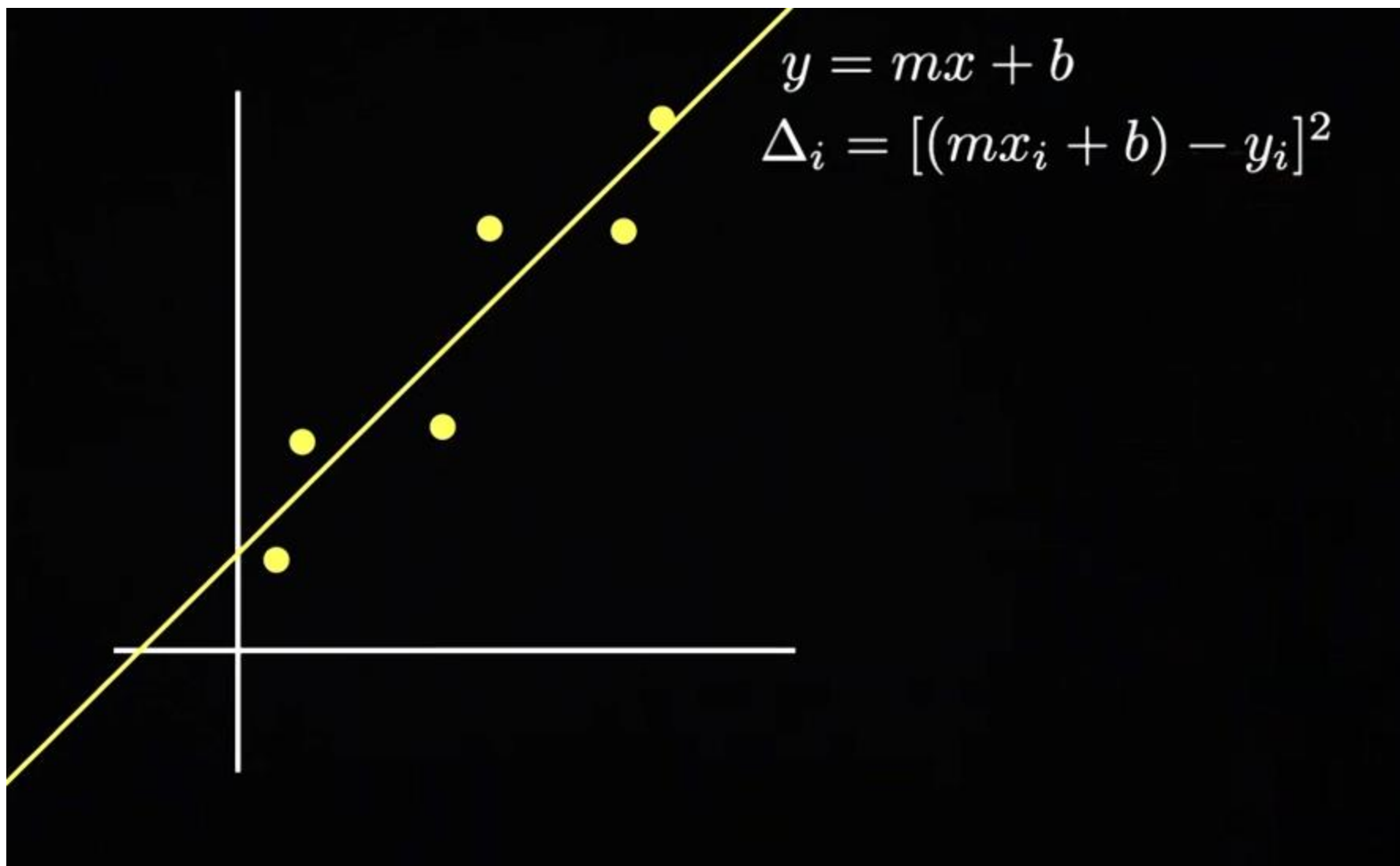
Regression
least-squares
line fitting
Parabola fitting

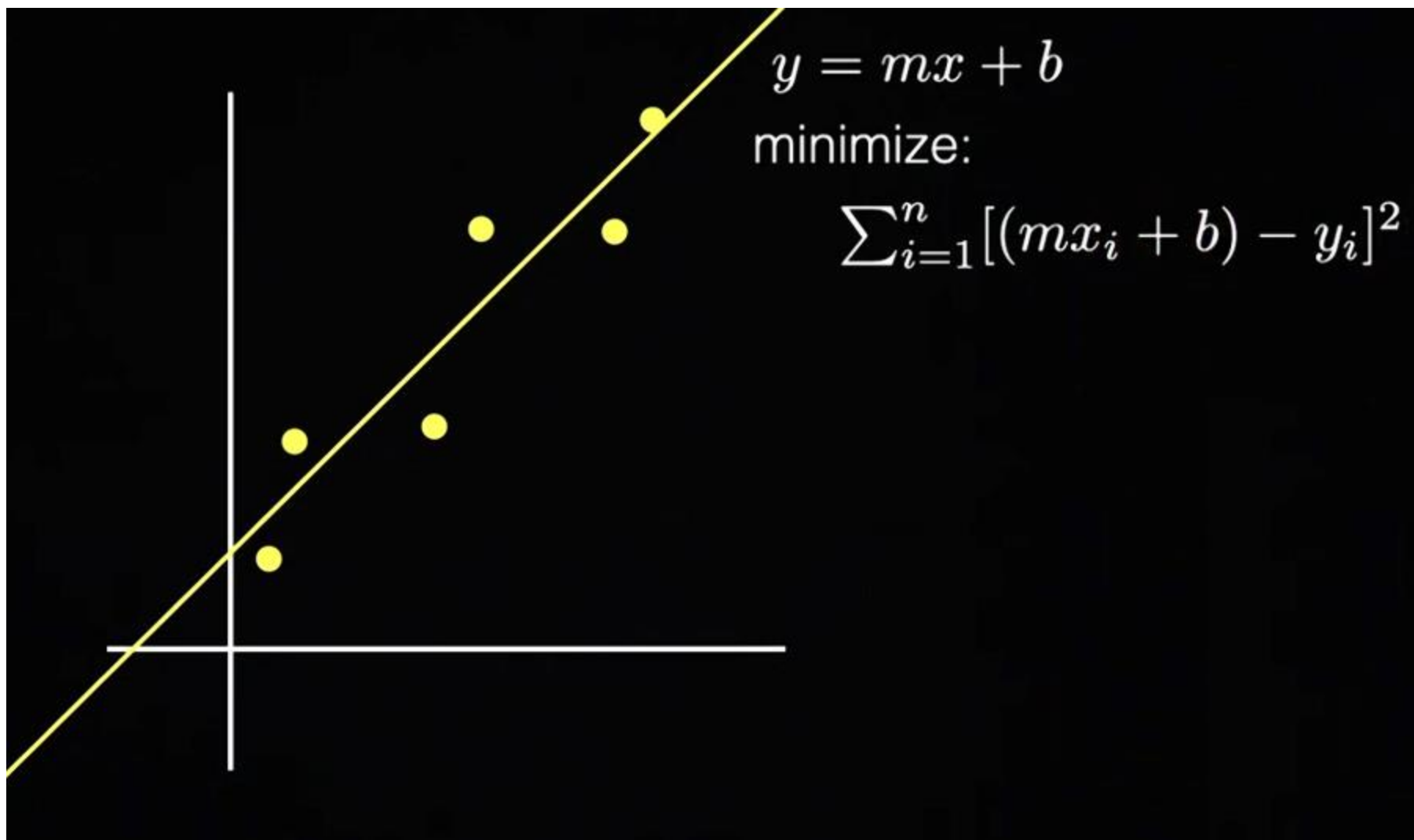


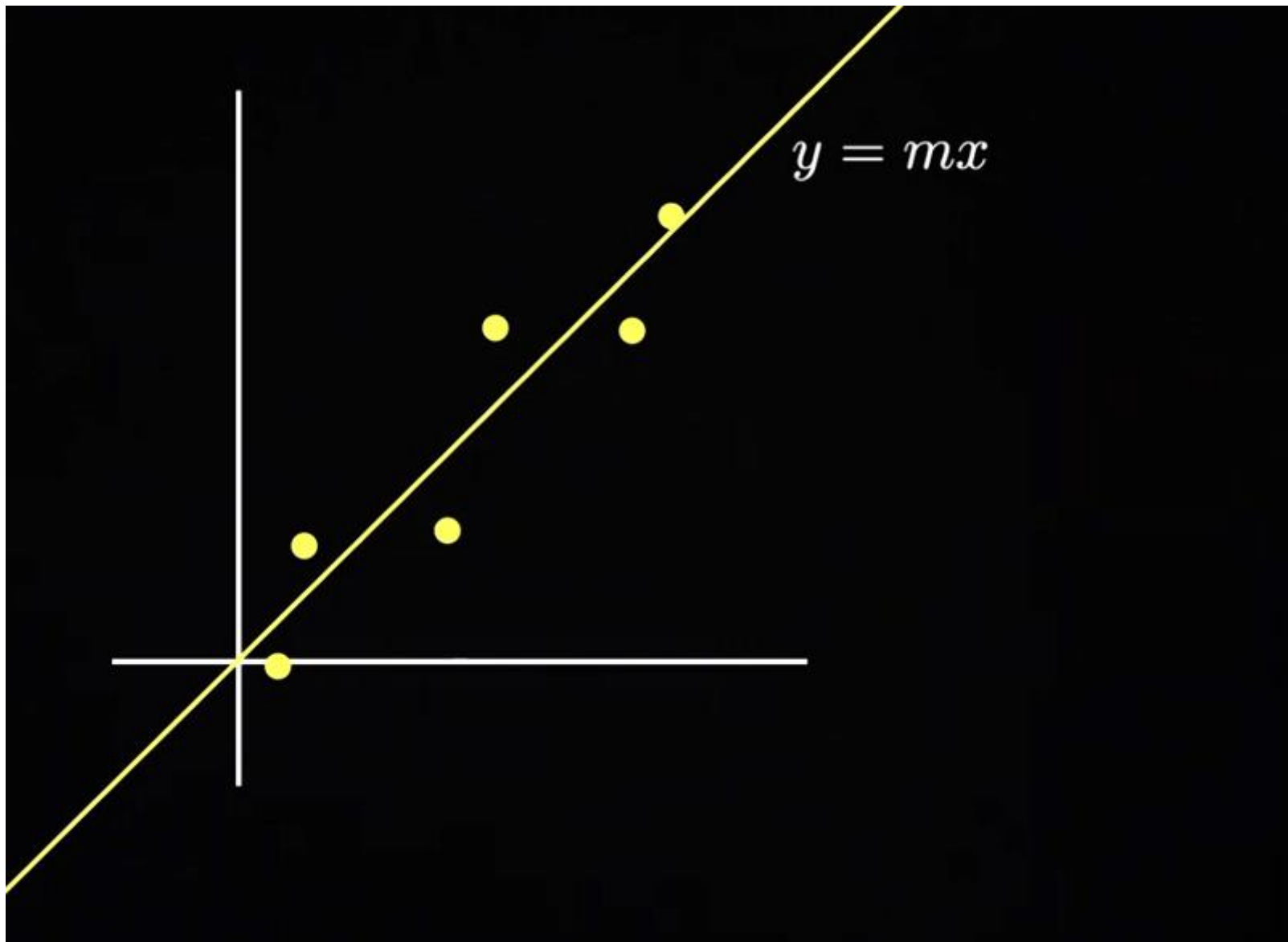


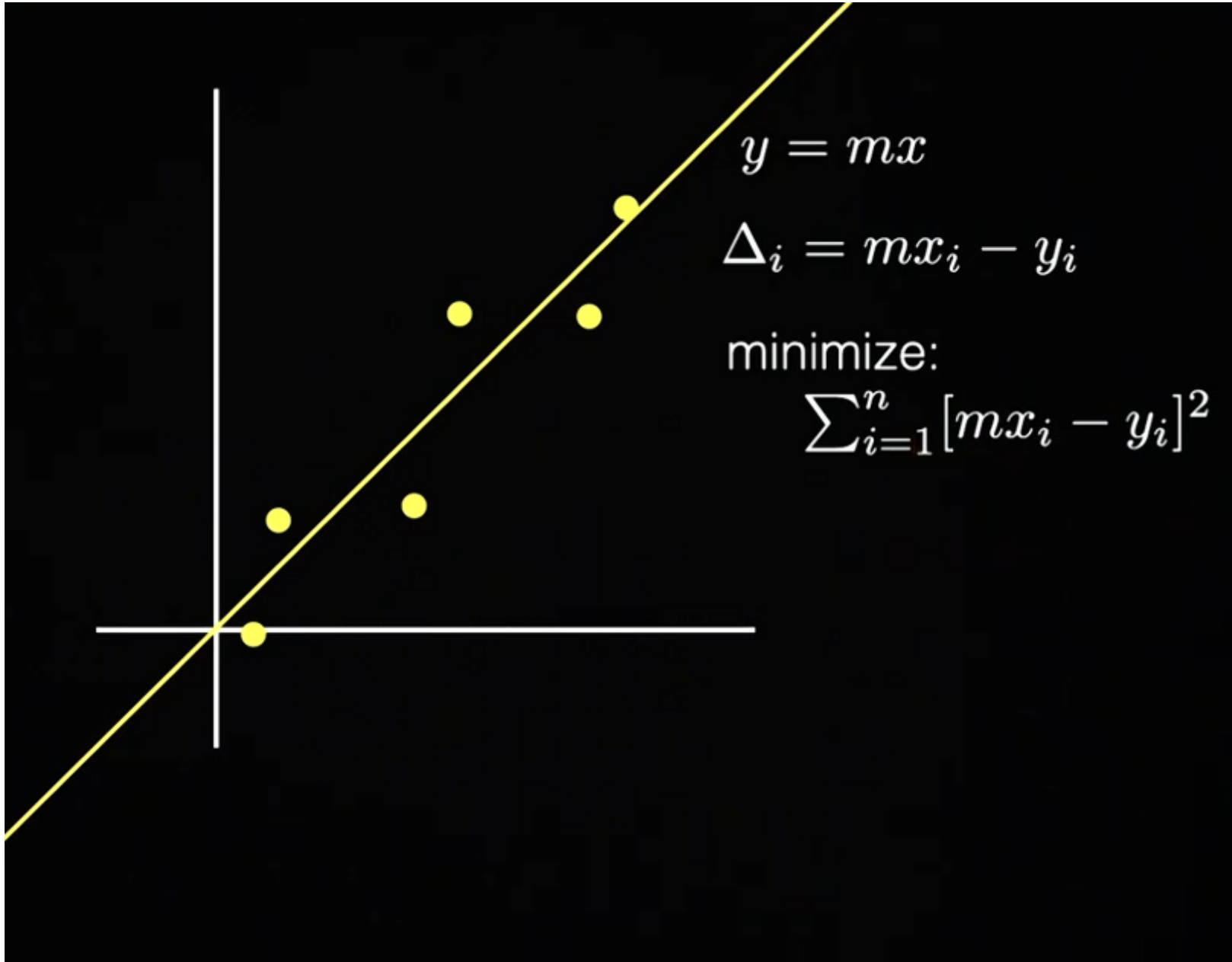












minimize:

$$\sum_{i=1}^n [mx_i - y_i]^2$$

$$\sum_{i=1}^n [(mx_i - y_i)(mx_i - y_i)]$$

$$\sum_{i=1}^n [m^2 x_i^2 - 2mx_i y_i + y_i^2]$$

$$m^2(x_1^2 + x_2^2 + \dots + x_n^2) - 2m(x_1 y_1 + x_2 y_2 + \dots + x_n y_n) + (y_1^2 + y_2^2 + \dots + y_n^2)$$

$$am^2 + bm + c$$



find m that minimizes sum of squared distances

$$E(m) = \sum_{i=1}^n [mx_i - y_i]^2$$

find m that minimizes sum of squared distances

$$E(m) = \sum_{i=1}^n [mx_i - y_i]^2$$

$$E(m) = am^2 + bm + c$$

$$\frac{dE(m)}{dm} = 0$$

$$\frac{dE(m)}{dm} = 2am + b$$

$$2am + b = 0$$

$$m = -\frac{b}{2a}$$

$$m = -\frac{b}{2a} = \frac{x_1 y_1 + \dots + x_n y_n}{x_1^2 + \dots + x_n^2}$$

find m and b that minimizes sum of squared distances

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$

find m and b that minimizes sum of squared distances

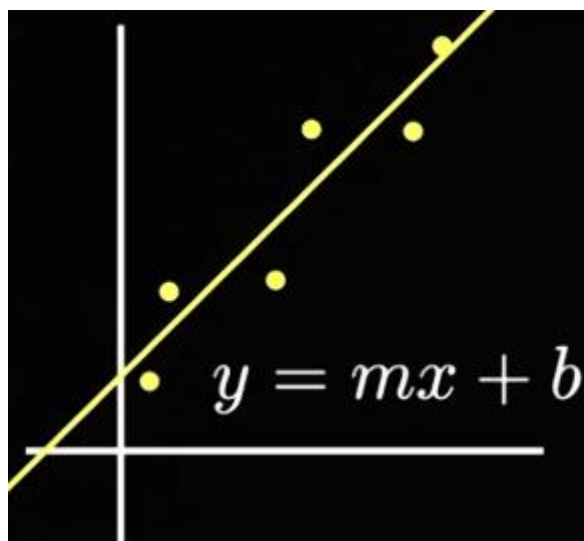
$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i][(mx_i + b) - y_i]$$

$$E(m, b) = \sum_{i=1}^n m^2 x_i^2 + 2mbx_i + b^2 - 2mx_i y_i - 2by_i + y_i^2$$

$$\frac{\partial E}{\partial m} = 0 \quad \text{and} \quad \frac{\partial E}{\partial b} = 0$$





$$\Delta_i = (mx_i + b) - y_i$$

$$\Delta_i = [(mx_i + b) - y_i]^2$$

minimize:

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$

$$mx_1 + b - y_1$$

$$mx_2 + b - y_2$$

...

$$mx_3 + b - y_3$$

minimize:

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$

$$mx_1 + b - y_1$$

$$mx_2 + b - y_2$$

...

$$mx_3 + b - y_3$$

$$\begin{pmatrix} x_1 & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} - y_1$$

$$\begin{pmatrix} x_2 & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} - y_2$$

...

$$\begin{pmatrix} x_n & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} - y_n$$

$$\begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} - \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

minimize:

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$



$$\sum_{i=1}^n \Delta_i^2$$

$$\sum_{i=1}^n \Delta_i^2 = (\Delta_1 \quad \Delta_2 \quad \cdots \quad \Delta_n) \begin{pmatrix} \Delta_1 \\ \Delta_2 \\ \vdots \\ \Delta_n \end{pmatrix}$$

$$\begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} - \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$X\vec{u} - \vec{y} = \vec{\Delta}$$

minimize:

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$



$$\sum_{i=1}^n \Delta_i^2$$

$$\sum_{i=1}^n \Delta_i^2 = (\Delta_1 \quad \Delta_2 \quad \cdots \quad \Delta_n) \begin{pmatrix} \Delta_1 \\ \Delta_2 \\ \vdots \\ \Delta_n \end{pmatrix}$$

$$\sum_{i=1}^n \Delta_i^2 = \vec{\Delta}^T \vec{\Delta}$$

$$\sum_{i=1}^n \Delta_i^2 = \|\vec{\Delta}\|^2$$

$$\|\vec{\Delta}\| = \sqrt{\Delta_1^2 + \Delta_2^2 + \cdots + \Delta_n^2}$$

$$\begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} - \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$X\vec{u} - \vec{y} = \vec{\Delta}$$

$$E(\vec{u}) = \|X\vec{u} - \vec{y}\|^2$$

minimize:

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$



$$E(\vec{u}) = \|X\vec{u} - \vec{y}\|^2$$

$$\frac{dE}{d\vec{u}} = \left(\frac{\partial E}{\partial m} \quad \frac{\partial E}{\partial b} \right)^T = 2X^T(X\vec{u} - \vec{y}) = 0$$

minimize:

$$E(m, b) = \sum_{i=1}^n [(mx_i + b) - y_i]^2$$



$$E(\vec{u}) = \|X\vec{u} - \vec{y}\|^2$$

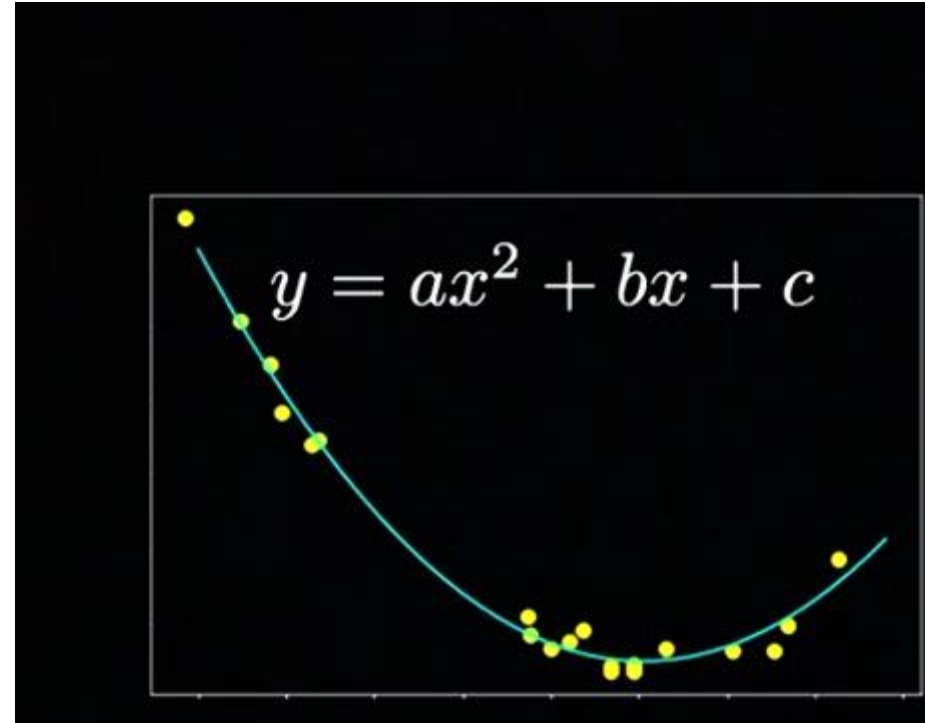
$$\frac{dE}{d\vec{u}} = \left(\frac{\partial E}{\partial m} \quad \frac{\partial E}{\partial b} \right)^T = 2X^T(X\vec{u} - \vec{y}) = 0$$

$$X^T X \vec{u} - X^T \vec{y} = 0$$

$$X^T X \vec{u} = X^T \vec{y}$$

$$\vec{u} = (X^T X)^{-1} X^T \vec{y}$$

Parabola Fitting



$$\Delta_i = [(ax_i^2 + bx_i + c) - y_i]^2$$

$$E(a, b, c) = \sum_{i=1}^n [(ax_i^2 + bx_i + c) - y_i]^2$$

$$\Delta_i = [(ax_i^2 + bx_i + c) - y_i]^2$$

$$E(a, b, c) = \sum_{i=1}^n [(ax_i^2 + bx_i + c) - y_i]^2$$

$$E(a, b, c) = \begin{pmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ \vdots & \vdots & \\ x_n^2 & x_n & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} - \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$\Delta_i = [(ax_i^2 + bx_i + c) - y_i]^2$$

$$E(a, b, c) = \sum_{i=1}^n [(ax_i^2 + bx_i + c) - y_i]^2$$

$$E(a, b, c) = \begin{pmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ \vdots & \vdots & \\ x_n^2 & x_n & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} - \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$E(\vec{u}) = \|X\vec{u} - \vec{y}\|^2$$

$$\Delta_i = [(ax_i^2 + bx_i + c) - y_i]^2$$

$$E(a, b, c) = \sum_{i=1}^n [(ax_i^2 + bx_i + c) - y_i]^2$$

$$E(a, b, c) = \begin{pmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ \vdots & \vdots & \\ x_n^2 & x_n & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} - \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$E(\vec{u}) = \|X\vec{u} - \vec{y}\|^2$$

$$\frac{dE}{d\vec{u}} = \left(\frac{\partial E}{\partial a} \quad \frac{\partial E}{\partial b} \quad \frac{\partial E}{\partial c} \right)^T = 2X^T(X\vec{u} - \vec{y}) = 0$$

$$X^T X \vec{u} - X^T \vec{y} = 0$$

$$\vec{u} = (X^T X)^{-1} X^T \vec{y}$$

Reference

- <https://www.youtube.com/watch?v=4cMqScDcsiU&list=PLhwIOYE-ldwL6h-peJADfNm8bbO3GIKEy&index=46>
- <https://www.youtube.com/watch?v=sgwxNXMEiz4&list=PLhwIOYE-ldwL6h-peJADfNm8bbO3GIKEy&index=47>
- https://www.youtube.com/watch?v=0rWtl_omFEE&list=PLhwIOYE-ldwL6h-peJADfNm8bbO3GIKEy&index=48